Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/11/2022
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Probing the Effect of Simulated Microgravity on the Pathogenic Potential of Cariogenic Streptococcus mutans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/25/2022 per NSSC information (Ed., 4/27/22).		
Key Personnel Changes/Previous PI:	August 2021: No key personnel changes to report.		
COI Name (Institution):	Shields, Robert Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC18K1463		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Streptococcus mutans is a member of the human oral microbiota and a major etiological agent of dental caries (cavities) due to its ability to form biofilm, adapt to oral cavity stresses, and rapidly ferment carbohydrates. These characteristics promote its predomination of dental plaque biofilm and combined with repeated cycles of fermentative acid production, leads to tooth demineralization and cavity formation. Using High Aspect Rotating Vessels (HARVs) in the low-shear modelled microgravity (LSMMG) model, we previously demonstrated that S. mutans displays several physiological changes in response to simulated microgravity, including altered cell aggregation, enhanced sensitivity to oxidative stress, and global alterations in gene expression. Furthermore, simulated microgravity exposure has been shown by others to cause increased mandibular and alveolar bone loss and decreased saliva flow, factors that could predispose astronauts to caries. Therefore, the goal of this application is to characterize the physiological and genetic response of S. mutans to simulated microgravity, focusing specifically on biofilm development, stress resistance, and identifying key gene(s) and/or the core transcriptomic response that favors S. mutans survival during microgravity growth. This will be accomplished by three objectives: (1) S. mutans will be cultured in biofilm-promoting media under simulated microgravity conditions using the LSMMG and random-positioning machine (RPM) ground-based models of growth. Changes in cell aggregation, stress resistance, and gene expression profiling (via RNAseq) will be assessed with respect to normal gravity control cultures, and these results will be compared using bioinformatics and statistical techniques. This analysis will allow us to make more refined predictions as to which physiological and gene expression changes are likely to occur in S. mutans during spaceflight, and how this may impact its pathogenic potential; (2) Changes in S. mutans biofilm development in response to simulated microgravity will be studied using 2 ground-based models (LSMMG, RPM). S. mutans will be assessed in these models for biofilm viability and ultrastructure, exopolysaccharide production, and for its ability to compete with non-cariogenic oral streptococci; (3) A S. mutans Tn-seq library will be screened for mutants that lose competitive fitness during LSMMG and RPM-based simulated microgravity growth. This will be accomplished by using high-throughput DNA sequencing of pooled batches of Tn mutants pre- and post-HARV growth under both simulated microgravity and normal gravity conditions. This research is directed at understanding the physiological and genetic response of S. mutans to microgravity, especially with respect to multicellular behavior (biofilm formation), stress resistance, and the genes that favor S. mutans survival during microgravity growth. Therefore, this application directly addresses guiding questions MB-1 (What underlying genetic, molecular and biochemical processes are influenced by the space environment?), MB-3 (How does spaceflight influence microbial biofilm formation and microbial community development?), and MB-4 (What are the mechanisms that effect changes such as the altered virulence or altered drug resistance observed in some organisms during spaceflight?) of the 2016-2025 Space Biology Science Plan, each of which aligns with Priority P1 (multigenerational studies of microbial populations) of the National Research Council's (NRC) 2011 Decadal Survey Report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Collectively, the research proposed in this application is directed at understanding the physiological and genetic response of S. mutans to simulated microgravity, especially with respect to multicellular behavior (biofilm formation), stress resistance, and the underlying genetics that promote S. mutans fitness and growth. It is anticipated that the collective data generated from these three objectives will allow us to make better predictions as to which physiological and gene expression changes are likely to occur in S. mutans during spaceflight, and how this may impact its disease-causing potential and effect on astronaut health. The data generated in objectives 1-3 of this application will also contribute to our understanding of the molecular mechanisms by which S. mutans and other oral bacteria contribute to dental caries (cavities), since the low-shear force experienced during HARV simulated microgravity growth is very similar to that of saliva flow in the oral cavity.
Task Progress:	In this fourth funding period, we re-grew Streptococcus mutans (S. mutans) under Rotary Cell Culture System (RCCS) normal gravity (NG), RCCS simulated microgravity (MG), and in High Aspect Rotating Vessels (HARVs) on a random positioning machine (RPM), to both mid-exponential (~3 hrs growth) and late-exponential growth phase (~6 hrs growth) (n=3 samples per condition per time point). RNA was isolated from cell pellets collected from all 18 samples, DNAse-treated, and was of high yield and integrity (A260/A280 ratios and Tapestation analysis/RIN). Ribodepletion, cDNA library construction, and Illumina High-Throughput sequencing was performed at the University of Florida Institute for Biotechnology Research (ICBR). Raw data was analyzed using Qiagen CLC Genomics Workbench (barcode trimming, in silico rRNA depletion, alignment of reads to S. mutans UA159 genome, and differential expression analysis of normalized reads). Bioinformatics and statistical analysis of the RNAseq differential expression (DE) analyses are ongoing. TnSeq RPM sample growth was attempted twice, but for some unknown reason the culture did not grow well in this model, even though the corresponding control (1 g) culture grew well. The PhD graduate student assigned to this project presented a poster on this work at the 2021 ASGSR conference. The Principal Investigator (PI) on this project has been invited to give an oral presentation on the Aim 2 RPM biofilm model at the 2022 Streptococcal Gordon Conference.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Abstracts for Journals and Proceedings	Davis KA, Rice KC. "Direct imaging of dual-species Streptococcus mutans & Streptococcus gordonii biofilms grown under simulated microgravity. " 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021 Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021